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TREND-BASED MOCK PAPER

GATE Instrumentation Engineering

Mock Paper — 04

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Instrumentation Engineering
Paper:	Mock 04
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

Unofficial practice material. Not affiliated with or endorsed by GATE/IIT.
Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

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Section A: General Aptitude

Q1. [1-MCQ] [1 mark]

20% of 200:

- (A) 20
- (B) 40
- (C) 80
- (D) 100

Section B: Instrumentation Engineering

Q2. [1-MCQ] [1 mark]

Determinant of 2x2 matrix:

- (A) $ad-bc$
- (B) $ad+bc$
- (C) $a+d$
- (D) ab

Q3. [1-MCQ] [1 mark]

Number of spanning trees in complete graph K_4 :

- (A) 4
- (B) 8
- (C) 16
- (D) 12

Q4. [1-MCQ] [1 mark]

Solution of $y'' + 4y = 0$ with $y(0)=1$, $y'(0)=0$, $y(\pi/2)$ equals:

- (A) 1
- (B) 0
- (C) -1

(D) $\cos(\pi)$

Q5. [1-MCQ] [1 mark]

If $f(x) = x^3 - 3x + 2$, number of real roots:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q6. [1-MCQ] [1 mark]

For a Poisson distribution with mean 4, variance is:

- (A) 4
- (B) 2
- (C) 16
- (D) 1

Q7. [2-MSQ] [2 marks]

$P(|Z| \leq 1)$ for $Z \sim N(0,1)$ is approximately:

- (A) 0.68
- (B) 0.32
- (C) 0.95
- (D) 0.16

Q8. [2-MCQ] [2 marks]

Gauss-Seidel method converges when the coefficient matrix is:

- (A) Symmetric
- (B) Positive definite
- (C) Diagonally dominant
- (D) Non-singular

Q9. [2-NAT] [2 marks]

Which of the following is a probability generating function for $\text{Poisson}(\lambda)$?

(Numerical Answer Type — Enter the numerical value)

Q10. [2-MCQ] [2 marks]

If $f(z) = z^2 + 3z + 2$ is analytic, then $\text{Re}(f(i)) =$

- (A) -1
- (B) -2
- (C) 1
- (D) 2

Q11. [2-NAT] [2 marks]

Laplace transform of $e^{-3t}u(t)$ is:

(Numerical Answer Type — Enter the numerical value)

Q12. [2-MSQ] [2 marks]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

- (A) 0.2
- (B) 0.5
- (C) 0.6
- (D) 0.4

Q13. [2-NAT] [2 marks]

The area of a circle of radius r is:

(Numerical Answer Type — Enter the numerical value)

Q14. [2-MCQ] [2 marks]

Consider $f(x) = |x|$ on $[-1,1]$. This function is:

- (A) Differentiable everywhere
 - (B) Continuous but not differentiable at 0
 - (C) Neither continuous nor differentiable
 - (D) Differentiable at 0 only
-

Q15. [2-NAT] [2 marks]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

(Numerical Answer Type — Enter the numerical value)

Q16. [2-MCQ] [2 marks]

The differential equation $y \, dx - x \, dy = 0$ has solution:

- (A) $y = C \cdot x$
 - (B) $y = C/x$
 - (C) $y = C \cdot x^2$
 - (D) $x^2 + y^2 = C$
-

Q17. [2-NAT] [2 marks]

The vector $(1,2,3)$ projected onto $(2,1,0)$ is:

(Numerical Answer Type — Enter the numerical value)

Q18. [1-NAT] [1 mark]

4-bit ripple adder, 4 FAs at 20ns each. Delay:

(Numerical Answer Type — Enter the numerical value)

Q19. [1-NAT] [1 mark]

Mod-12 counter minimum FFs:

(Numerical Answer Type — Enter the numerical value)

Q20. [1-NAT] [1 mark]

4:1 MUX select lines:

(Numerical Answer Type — Enter the numerical value)

Q21. [2-NAT] [2 marks]

Race-around in JK FF when:

(Numerical Answer Type — Enter the numerical value)

Q22. [2-NAT] [2 marks]

One's complement of 11010110:

(Numerical Answer Type — Enter the numerical value)

Q23. [2-MCQ] [2 marks]

Half adder outputs:

- (A) Sum and carry

- (B) Sum only
- (C) Carry only
- (D) Difference

Q24. [2-MCQ] [2 marks]

Setup time violation: solution:

- (A) Decrease clock freq
- (B) Increase clock freq
- (C) Use latch
- (D) Both A and C

Q25. [2-MSQ] [2 marks]

2K x 8 ROM address lines:

- (A) 2
- (B) 8
- (C) 11
- (D) 2048

Q26. [2-MCQ] [2 marks]

4-bit Johnson counter states:

- (A) 4
- (B) 8
- (C) 16
- (D) 10

Q27. [1-NAT] [1 mark]

PID controller has:

(Numerical Answer Type — Enter the numerical value)

Q28. [1-MCQ] [1 mark]

$s^2 + 2s + 5 = 0$. Damping ratio:

- (A) 0.2
- (B) 0.447
- (C) 0.707
- (D) 1.0

Q29. [1-MCQ] [1 mark]

Gain margin is defined at:

- (A) Phase crossover
- (B) Gain crossover
- (C) Both
- (D) Neither

Q30. [2-NAT] [2 marks]

Integral controller eliminates:

(Numerical Answer Type — Enter the numerical value)

Q31. [2-NAT] [2 marks]

Root locus branches start at poles and end at:

(Numerical Answer Type — Enter the numerical value)

Q32. [2-MCQ] [2 marks]

Nyquist criterion: $N = Z - P$. If $P=1$, $N=-1$:

- (A) 1 zero in RHP
- (B) Stable
- (C) 1 clockwise encirclement needed
- (D) Unstable

Q33. [2-MCQ] [2 marks]

A system with characteristic $1+s = 0$ has:

- (A) Stable
- (B) Marginally stable
- (C) Unstable
- (D) Cannot determine

Q34. [2-NAT] [2 marks]

Type 1 system with $K_v=5$. Steady-state error for unit ramp:

(Numerical Answer Type — Enter the numerical value)

Q35. [1-MCQ] [1 mark]

CMRR of diff amp depends on:

- (A) Only R_C
- (B) Only β
- (C) Active load matching
- (D) Temperature

Q36. [1-MCQ] [1 mark]

CC (emitter follower) has:

- (A) High voltage gain
- (B) High current gain
- (C) Low input Z
- (D) Voltage gain > 1

Q37. [1-NAT] [1 mark]

Class B push-pull efficiency:

(Numerical Answer Type — Enter the numerical value)

Q38. [2-MCQ] [2 marks]

CE amp $h_{fe}=100$, $h_{ie}=1k$, $R_C=2k$. Voltage gain:

- (A) -200
- (B) -100
- (C) -150
- (D) -50

Q39. [2-MCQ] [2 marks]

Op-amp integrator $R=10k$, $C=1\mu F$. Unit step after 1ms:

- (A) -0.1V
- (B) -0.01V
- (C) -0.001V
- (D) -1V

Q40. [2-MCQ] [2 marks]

Frequency response RC integrator (high-pass) corner:

- (A) $1/(2\pi RC)$
- (B) RC
- (C) $1/RC$
- (D) $2\pi RC$

Q41. [2-MCQ] [2 marks]

RC low-pass at cutoff, gain:

- (A) 0 dB
- (B) -3 dB
- (C) -20 dB
- (D) -40 dB/dec

Q42. [2-MCQ] [2 marks]

Input offset voltage causes:

- (A) Output saturates
- (B) Output DC offset
- (C) No effect
- (D) BW increases

Q43. [1-MCQ] [1 mark]

Wheatstone bridge null condition:

- (A) $R_1/R_2 = R_3/R_4$
- (B) $R_1 R_4 = R_2 R_3$
- (C) Both A and B
- (D) $R_1 + R_2 = R_3 + R_4$

Q44. [1-MCQ] [1 mark]

$H(s) = (s^2 + 4s + 3)/(s^2 + 2s + 2)$. Zeros at:

- (A) $s = -1, -3$
- (B) $s = -1 + j, -1 - j$
- (C) $s = -1, -3$ and $-1 + j$
- (D) $s = -2 + j, -2 - j$

Q45. [1-NAT] [1 mark]

Norton equivalent of Thevenin ($V_{th} = 10V$, $R_{th} = 5\Omega$):

(Numerical Answer Type — Enter the numerical value)

Q46. [1-NAT] [1 mark]

Time constant of RL ($R = 10\Omega$, $L = 5H$):

(Numerical Answer Type — Enter the numerical value)

Q47. [2-MCQ] [2 marks]

Thevenin equivalent with $V_{th} = 10V$, $R_{th} = 5\Omega$, load = 5Ω . Output:

- (A) 5V, 1A
- (B) 5V, 0.5A
- (C) 2.5V, 0.5A
- (D) 10V, 1A

Q48. [2-MCQ] [2 marks]

Laplace transform of $e^{(-3t)}u(t)$:

- (A) $1/(s+3)$
- (B) $1/(s-3)$
- (C) $1/s$
- (D) e^{-3}/s

Q49. [2-MCQ] [2 marks]

Two-port h-parameter h_{11} has units:

- (A) Ohm
- (B) Mho
- (C) Volt
- (D) Dimensionless

Q50. [2-NAT] [2 marks]

Y-parameter Y_{11} for series impedance Z is:

(Numerical Answer Type — Enter the numerical value)

Q51. [1-NAT] [1 mark]

Carson rule: $\Delta f = 75\text{kHz}$, $f_m = 15\text{kHz}$. BW:

(Numerical Answer Type — Enter the numerical value)

Q52. [1-MCQ] [1 mark]

QPSK transmits bits per symbol:

- (A) 1
- (B) 2
- (C) 4
- (D) 8

Q53. [2-NAT] [2 marks]

AM signal carrier 1kW, modulation 0.5. Total power:

(Numerical Answer Type — Enter the numerical value)

Q54. [2-MCQ] [2 marks]

NBFM bandwidth approx:

- (A) $2f_m$
- (B) $2(\Delta f + f_m)$
- (C) Δf
- (D) f_m only

Q55. [2-MCQ] [2 marks]

Shannon capacity: BW=1MHz, SNR=15. Capacity:

- (A) 4 Mbps
- (B) 1 Mbps
- (C) 15 Mbps
- (D) 5 Mbps

Q56. [2-NAT] [2 marks]

PCM $L=16$ levels. Bits per sample:

(Numerical Answer Type — Enter the numerical value)

Q57. [1-MCQ] [1 mark]

Sum of first n odd numbers:

- (A) n^2
- (B) $2n$
- (C) $n(n+1)$
- (D) $n(n-1)$

Q58. [1-MCQ] [1 mark]

Compound interest: P at r% for 2 years =

- (A) $P(1+r/100)^2$
- (B) $P(1+2r/100)$
- (C) $P + Pr/100 \times 2$
- (D) $P \times r/100$

Q59. [1-MCQ] [1 mark]

If $x + 1/x = 3$, then $x^3 + 1/x^3 =$

- (A) 18
- (B) 27
- (C) 9
- (D) 3

Q60. [1-NAT] [1 mark]

8-bit ADC has resolution:

(Numerical Answer Type — Enter the numerical value)

Q61. [1-MCQ] [1 mark]

Instrumentation amp has:

- (A) 1 op-amp
- (B) 2
- (C) 3
- (D) 4

Q62. [1-MCQ] [1 mark]

FM uses:

- (A) Amplitude
- (B) Frequency
- (C) Phase
- (D) None

Q63. [1-MCQ] [1 mark]

Closed-loop uses:

- (A) No feedback
- (B) Feedback
- (C) Feedforward
- (D) None

Q64. [1-MCQ] [1 mark]

LVDT measures:

- (A) Displacement
- (B) Velocity
- (C) Acceleration

(D) Force

Q65. [1-NAT] [1 mark]

Moving coil instrument measures:

(Numerical Answer Type — Enter the numerical value)

ANSWER KEY

Q. No.	Type	Answer	Marks
1	1MCQ	B	1
2	1MCQ	A	1
3	1MCQ	A	1
4	1NAT	C	1
5	1NAT	A	1
6	1MCQ	C	1
7	1MCQ	B	1
8	1MCQ	B	1
9	1NAT	B	1
10	1MCQ	C	1
11	1MCQ	C	1
12	1NAT	A	1
13	1NAT	C	1

14	1MCQ	C	1
15	1MCQ	A	1
16	1MCQ	C	1
17	1MCQ	B	1
18	1MCQ	B	1
19	1MCQ	A	1
20	1NAT	B	1
21	1MCQ	A	1
22	1MCQ	B	1
23	1NAT	B	1
24	1NAT	A	1
25	1NAT	C	1
26	1MCQ	A	1
27	1MCQ	A	1
28	1MCQ	B	1

29	1MCQ	A	1
30	1NAT	B	1
31	2MCQ	A	2
32	2NAT	B	2
33	2NAT	A	2
34	2NAT	A	2
35	2MCQ	A	2
36	2MCQ	A	2
37	2MCQ	A	2
38	2MCQ	A	2
39	2MCQ	A	2
40	2NAT	B	2
41	2MSQ	A	2
42	2MCQ	A	2
43	2NAT	C	2

44	2MCQ	C	2
45	2MCQ	C	2
46	2MCQ	A	2
47	2NAT	A	2
48	2MCQ	A	2
49	2MCQ	A	2
50	2NAT	A	2
51	2MCQ	A	2
52	2MCQ	B	2
53	2MSQ	B	2
54	2MSQ	C	2
55	2NAT	A	2
56	2NAT	D	2
57	2MCQ	B	2
58	2MCQ	B	2

59	2NAT	B	2
60	2MCQ	B	2
61	2MCQ	B	2
62	2MCQ	B	2
63	2NAT	B	2
64	2NAT	B	2
65	2NAT	B	2

DETAILED SOLUTIONS

Q1.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Quantitative | Difficulty: moderate

Q2.

Answer: A

Solution:

Correct Answer: A

Subject: Engineering Mathematics | Topic: Linear Algebra | Difficulty: easy

Q3.

Answer: A

Solution:

Correct Answer: A

Subject: Sensors & Instrumentation | Topic: Transducers | Difficulty: difficult

Q4.

Answer: C

Solution:

Correct Answer: C

Subject: Control Systems | Topic: Controllers | Difficulty: moderate

Q5.

Answer: A

Solution:

Correct Answer: A

Subject: Analog & Digital Electronics | Topic: ADC | Difficulty: moderate

Q6.

Answer: C

Solution:

Correct Answer: C

Subject: Analog & Digital Electronics | Topic: Op-Amps | Difficulty: difficult

Q7.

Answer: B

Solution:

Correct Answer: B

Subject: Communications & Process Control | Topic: Modulation | Difficulty: difficult

Q8.

Answer: B

Solution:

Correct Answer: B

Subject: Communications & Process Control | Topic: Control Loops | Difficulty: moderate

Q9.

Answer: B

Solution:

Correct Answer: B

Subject: Measurement Systems | Topic: Instrumentation | Difficulty: easy

Q10.

Answer: C

Solution:

Kirchhoff: for K_n , spanning trees = $n^{n-2} = 4^2 = 16$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q11.

Answer: C

Solution:

$y = A \cos(2x) + B \sin(2x)$. $A=1$, $B=0$. $y(\pi/2) = \cos(\pi) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q12.

Answer: A

Solution:

$BW = 2 \times (75+15) = 180 \text{ kHz}$.

Subject: Communication | Topic: Communication | Difficulty: moderate

Q13.

Answer: C

Solution:

Carry ripples through 4 FAs. $t_{pd} = 4 \times 20 = 80 \text{ ns}$.

Subject: Digital Electronics | Topic: Digital Electronics | Difficulty: moderate

Q14.

Answer: C

Solution:

CMRR depends on active load matching. Active load improves CMRR.

Subject: Analog Electronics | Topic: Analog Electronics | Difficulty: moderate

Q15.

Answer: A

Solution:

$1+3+5+\dots+(2n-1) = n^2$.

Subject: Signal Processing | Topic: Signal Processing | Difficulty: moderate

Q16.

Answer: C

Solution:

Null: $R_1/R_2 = R_3/R_4$, which implies $R_1 \times R_4 = R_2 \times R_3$.

Subject: Network Theory | Topic: Network Theory | Difficulty: difficult

Q17.

Answer: B

Solution:

$f(0)=2$, $f(1)=0$, $f(-1)=4$, $f(2)=4$. $f(-2)=-8+6+2=0$. Roots at $x=1$ (double?) and $x=-2$. $f(x)=(x-1)^2(x+2)$. Two distinct real roots.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q18.

Answer: B

Solution:

CC: $A_v \sim 1$, high input Z, high current gain, low output Z.

Subject: Analog Electronics | Topic: Analog Electronics | Difficulty: easy

Q19.

Answer: A

Solution:

$CI = P(1+r/100)^n$. For 2 years: $P(1+r/100)^2$.

Subject: Signal Processing | Topic: Signal Processing | Difficulty: moderate

Q20.

Answer: B

Solution:

$2^n \geq 12$. $n=4$ gives 16 states. Minimum 4 FFs.

Subject: Digital Electronics | Topic: Digital Electronics | Difficulty: moderate

Q21.

Answer: A

Solution:

Zeros from numerator: $s^2+4s+3 = (s+1)(s+3)$. Zeros at $s=-1$, -3 .

Subject: Network Theory | Topic: Network Theory | Difficulty: moderate

Q22.

Answer: B

Solution:

Compare to $s^2 + 2\zeta\omega_n s + \omega_n^2$. $2\zeta\omega_n \sqrt{5} = 2$. $\zeta = 2/(2\sqrt{5}) = 1/\sqrt{5}$ approx 0.447.

Subject: Control Systems | Topic: Control Systems | Difficulty: difficult

Q23.

Answer: B

Solution:

4:1: $\log_2(4) = 2$ select lines.

Subject: Digital Electronics | Topic: Digital Electronics | Difficulty: easy

Q24.

Answer: A

Solution:

$I_n = V_{th}/R_{th} = 10/5 = 2A$. $R_n = R_{th} = 5\Omega$.

Subject: Network Theory | Topic: Network Theory | Difficulty: difficult

Q25.

Answer: C

Solution:

Class B max efficiency: $\pi/4$ approx 78.5%.

Subject: Analog Electronics | Topic: Analog Electronics | Difficulty: moderate

Q26.

Answer: A

Solution:

Cube: $(x+1/x)^3 = x^3 + 3x + 3/x + 1/x^3 = 27$. $x^3 + 1/x^3 = 27 - 3 = 24$.

Subject: Signal Processing | Topic: Signal Processing | Difficulty: easy

Q27.

Answer: A

Solution:

Gain margin: at phase crossover frequency (phase = -180 deg), $1/|GH(j\omega)|$ in dB.

Subject: Control Systems | Topic: Control Systems | Difficulty: moderate

Q28.

Answer: B

Solution:

QPSK: 4 phase states. $\log_2(4) = 2$ bits per symbol.

Subject: Communication | Topic: Communication | Difficulty: easy

Q29.

Answer: A

Solution:

Poisson: mean = lambda = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q30.

Answer: B

Solution:

$\tau = L/R = 5/10 = 0.5$ s.

Subject: Network Theory | Topic: Network Theory | Difficulty: moderate

Q31.

Answer: A

Solution:

$A_v = -h_{fe}R_C/h_{ie} = -100 \times 2000/1000 = -200$.

Subject: Analog Electronics | Topic: Analog Electronics | Difficulty: moderate

Q32.

Answer: B

Solution:

Race-around: $J=K=1$ with clock pulse wider than propagation delay.

Subject: Digital Electronics | Topic: Digital Electronics | Difficulty: moderate

Q33.

Answer: A

Solution:

Flip all bits: 11010110 -> 00101001.

Subject: Digital Electronics | Topic: Digital Electronics | Difficulty: easy

Q34.

Answer: A

Solution:

$P_{\text{total}} = P_c(1+m^2/2) = 1000*(1+0.125) = 1125W = 1.125kW.$

Subject: Communication | Topic: Communication | Difficulty: easy

Q35.

Answer: A

Solution:

Half adder: Sum=A XOR B, Carry=A AND B.

Subject: Digital Electronics | Topic: Digital Electronics | Difficulty: moderate

Q36.

Answer: A

Solution:

$V_{\text{out}} = 10*5/(5+5) = 5V. I = 5/5 = 1A.$

Subject: Network Theory | Topic: Network Theory | Difficulty: moderate

Q37.

Answer: A

Solution:

$L\{e^{(-at)}\} = 1/(s+a).$ Here $a=3: 1/(s+3).$

Subject: Network Theory | Topic: Network Theory | Difficulty: moderate

Q38.

Answer: A

Solution:

NBFM: BW approx $2*f_m$ (same as AM). WBFM: Carson rule.

Subject: Communication | Topic: Communication | Difficulty: moderate

Q39.

Answer: A

Solution:

$h_{11} = (V_1/I_1)$ with output shorted. Units: $V/A = \text{Ohm}.$

Subject: Network Theory | Topic: Network Theory | Difficulty: moderate

Q40.

Answer: B

Solution:

Integral control: eliminates steady-state error for step/ramp inputs. Increases system type by 1.

Subject: Control Systems | Topic: Control Systems | Difficulty: moderate

Q41.

Answer: A

Solution:

Standard normal: $P(-1 < Z < 1)$ approx 0.68.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q42.

Answer: A

Solution:

$C = B*\log_2(1+SNR) = 1e6*\log_2(16) = 1e6*4 = 4 \text{ Mbps}.$

Subject: Communication | Topic: Communication | Difficulty: difficult

Q43.

Answer: C

Solution:

Root locus: starts at OL poles, ends at OL zeros (or infinity if $P > Z$).

Subject: Control Systems | Topic: Control Systems | Difficulty: moderate

Q44.

Answer: C

Solution:

Gauss-Seidel converges if matrix is diagonally dominant. Jacobi needs symmetric positive definite.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q45.

Answer: C

Solution:

$N = Z - P$. $N = -1$ means one clockwise encirclement of -1 point needed. $Z = N + P = -1 + 1 = 0$. Stable.

Subject: Control Systems | Topic: Control Systems | Difficulty: moderate

Q46.

Answer: A

Solution:

$V_{out} = -t/(RC) = -100 \cdot t$. At 1ms: -0.1V.

Subject: Analog Electronics | Topic: Analog Electronics | Difficulty: moderate

Q47.

Answer: A

Solution:

PGF of Poisson(λ): $G(z) = E[z^X] = \exp(\lambda(z-1))$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q48.

Answer: A

Solution:

Setup: data not stable before clock. Lower clock freq gives longer period.

Subject: Digital Electronics | Topic: Digital Electronics | Difficulty: moderate

Q49.

Answer: A

Solution:

$f(i) = i^2 + 3i + 2 = -1 + 3i$. $\text{Re}(f(i)) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q50.

Answer: A

Solution:

$L\{e^{-at}\} = 1/(s+a)$. Here $a=3$: $1/(s+3)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q51.

Answer: A

Solution:

RC high-pass cutoff: $1/(2\pi RC)$.

Subject: Analog Electronics | Topic: Analog Electronics | Difficulty: moderate

Q52.

Answer: B

Solution:

Root at $s = -1$ (on left half plane). Simple pole on negative real axis: stable (marginally, as $t \rightarrow \infty$).

Subject: Control Systems | Topic: Control Systems | Difficulty: easy

Q53.

Answer: B

Solution:

$P(A|B) = P(A \cap B)/P(B) = 0.2/0.4 = 0.5$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q54.

Answer: C

Solution:

$2K = 2048$ locations. $\log_2 = 11$ address lines.

Subject: Digital Electronics | Topic: Digital Electronics | Difficulty: easy

Q55.

Answer: A

Solution:

Area = πr^2 . Circumference = $2\pi r$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q56.

Answer: D

Solution:

Series Z: $I_1 = I_2$, $V_1 - V_2 = Z I_1$. $Y_{11} = I_1/V_1$ with $I_2 = 0$ impossible ($I_1 = I_2$). $Y_{11} = 0$ for pure series.

Subject: Network Theory | Topic: Network Theory | Difficulty: easy

Q57.

Answer: B

Solution:

Johnson (4-bit): 8 states.

Subject: Digital Electronics | Topic: Digital Electronics | Difficulty: easy

Q58.

Answer: B

Solution:

$|x|$ is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q59.

Answer: B

Solution:

$(1 + 1/n)^{2n} = [(1 + 1/n)^n]^2 \rightarrow e^2$ as $n \rightarrow \infty$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q60.

Answer: B

Solution:

At f_c : gain = $1/\sqrt{2}$ = -3 dB.

Subject: Analog Electronics | Topic: Analog Electronics | Difficulty: moderate

Q61.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C*x$. Wait: $y dx = x dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C*x$. That's option A. Hmm, let me check: $y dx - x dy = 0 \Rightarrow y dx = x dy \Rightarrow dy/y = dx/x \Rightarrow y = C*x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q62.

Answer: B

Solution:

Input offset: differential DC input causing DC output offset.

Subject: Analog Electronics | Topic: Analog Electronics | Difficulty: moderate

Q63.

Answer: B

Solution:

$L = 2^n$. $n = \log_2(16) = 4$ bits.

Subject: Communication | Topic: Communication | Difficulty: moderate

Q64.

Answer: B

Solution:

Type 1: $\text{ess} = 1/K_v = 1/5 = 0.2$ for unit ramp.

Subject: Control Systems | Topic: Control Systems | Difficulty: moderate

Q65.

Answer: B

Solution:

$\text{proj} = ((v \cdot w)/(w \cdot w)) * w = (4/5) * (2, 1, 0) = (8/5, 4/5, 0)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

PAPER BLUEPRINT

Subject Distribution:

Engineering Mathematics: 16 questions, 27 marks
Digital Electronics: 9 questions, 15 marks
Control Systems: 8 questions, 13 marks
Analog Electronics: 8 questions, 13 marks
Network Theory: 8 questions, 12 marks
Communication: 6 questions, 10 marks
Signal Processing: 3 questions, 3 marks
Analog & Digital Electronics: 2 questions, 2 marks
Communications & Process Control: 2 questions, 2 marks
General Aptitude: 1 questions, 1 marks
Sensors & Instrumentation: 1 questions, 1 marks
Measurement Systems: 1 questions, 1 marks

Difficulty Distribution:

Easy: 17 | Moderate: 40 | Difficult: 8

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.